

Technology Stack

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction using AI/ML)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build this solution. A well-chosen technology stack ensures that the application is scalable, maintainable, and aligned with project requirements.

The table below details the specific technologies chosen for each component of the architecture with justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3	Used to build the web interface — home page, input form, and result page. Simple and lightweight, suitable for a local Flask-based ML application.
2	Backend / Server-Side	Python 3.x, Flask	Python is used for ML model development. Flask is the web framework that handles routing, processes user inputs, and serves predictions. Lightweight and easy to integrate with ML models.
3	Database / Data Storage	CSV File (HDI.csv) Pickle (.pkl)	The HDI dataset is stored as a CSV file. The trained Linear Regression model is serialized using Pickle for persistent storage and reuse without retraining.
4	Cloud / Hosting / Deployment	Local Server (Flask Dev Server, Port 5000)	The application runs locally on the development machine using Flask's built-in server. No external cloud deployment is required for this academic project.
5	Version Control & CI/CD	GitHub	GitHub is used to host and version-control all project files including code, dataset, notebooks, and documentation. Enables team collaboration and mentor

			review.
6	Third-Party APIs / Other Tools	Kaggle (Dataset Source) Anaconda, VS Code, Jupyter Notebook NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn	Kaggle provides the HDI dataset. Anaconda manages the Python environment. VS Code and Jupyter Notebook are used for development. ML libraries handle data processing, visualization, and model training.